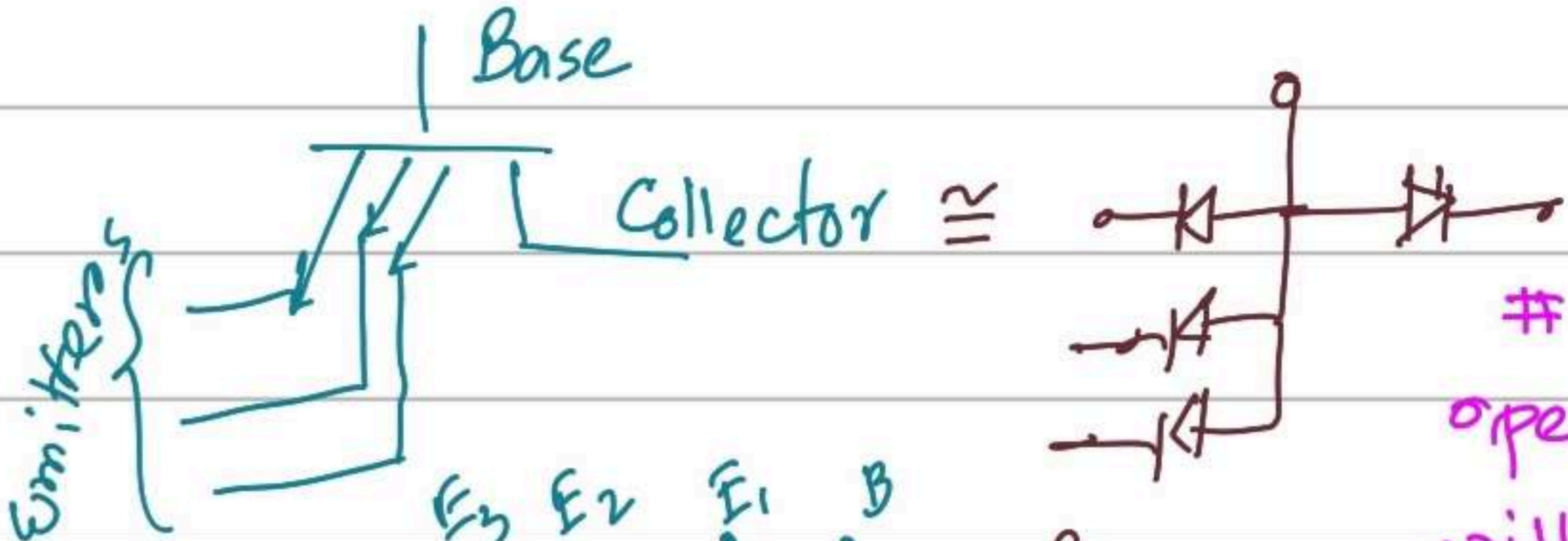
"Transistor-Transistor Logic"

Bipolar, → Saturated → TTL
BJT

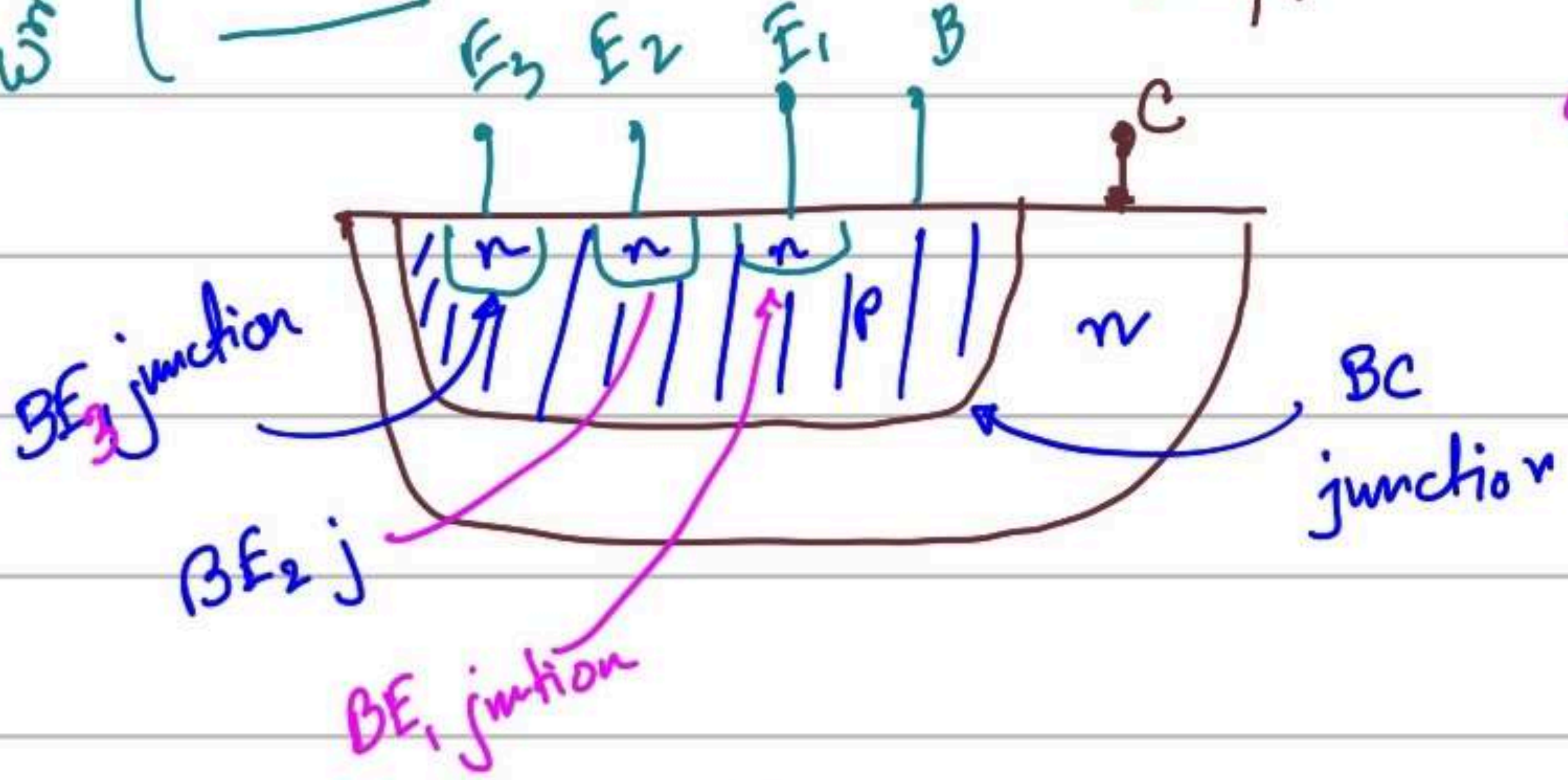
Switching time is faster than DTL.

The input terminals are connected to the emitter terminal of a transistor and the output terminal is connected to a switching transistor.

input terminals will be connected to multi-emitter transistor.



Transistor's operating mode will be determined from three emitter inputs altogether.



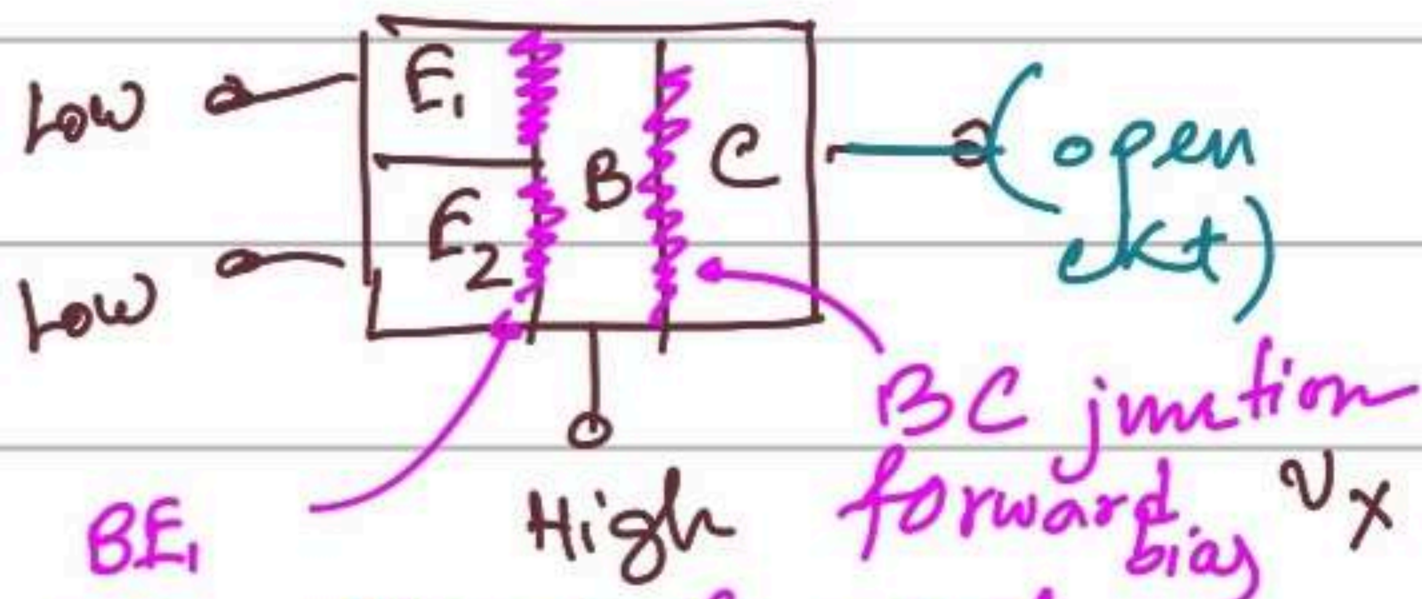
Basic Operation:

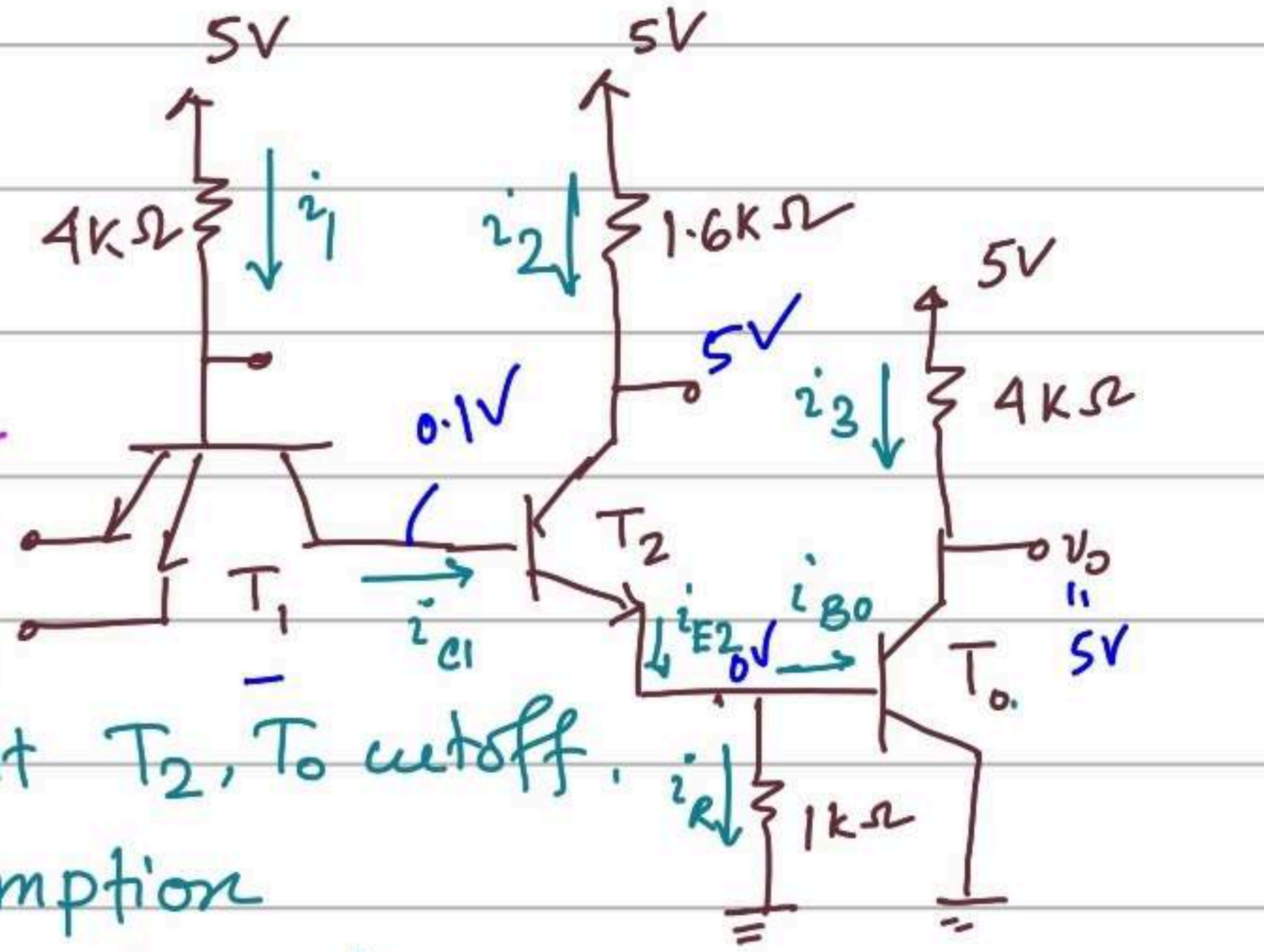
find out the current and voltage of the following gate. Verify that it works like a NAND gate.

Case ①. $v_X = v_Y = 0.1V$ [Neaman] $v_{CE}(sat) = 0.1V$

$v_{BE}(sat) = 0.8V$

For T_1 transistor





BE_2 will be in forward bias.

Our assumption is that T_2, T_o cutoff.

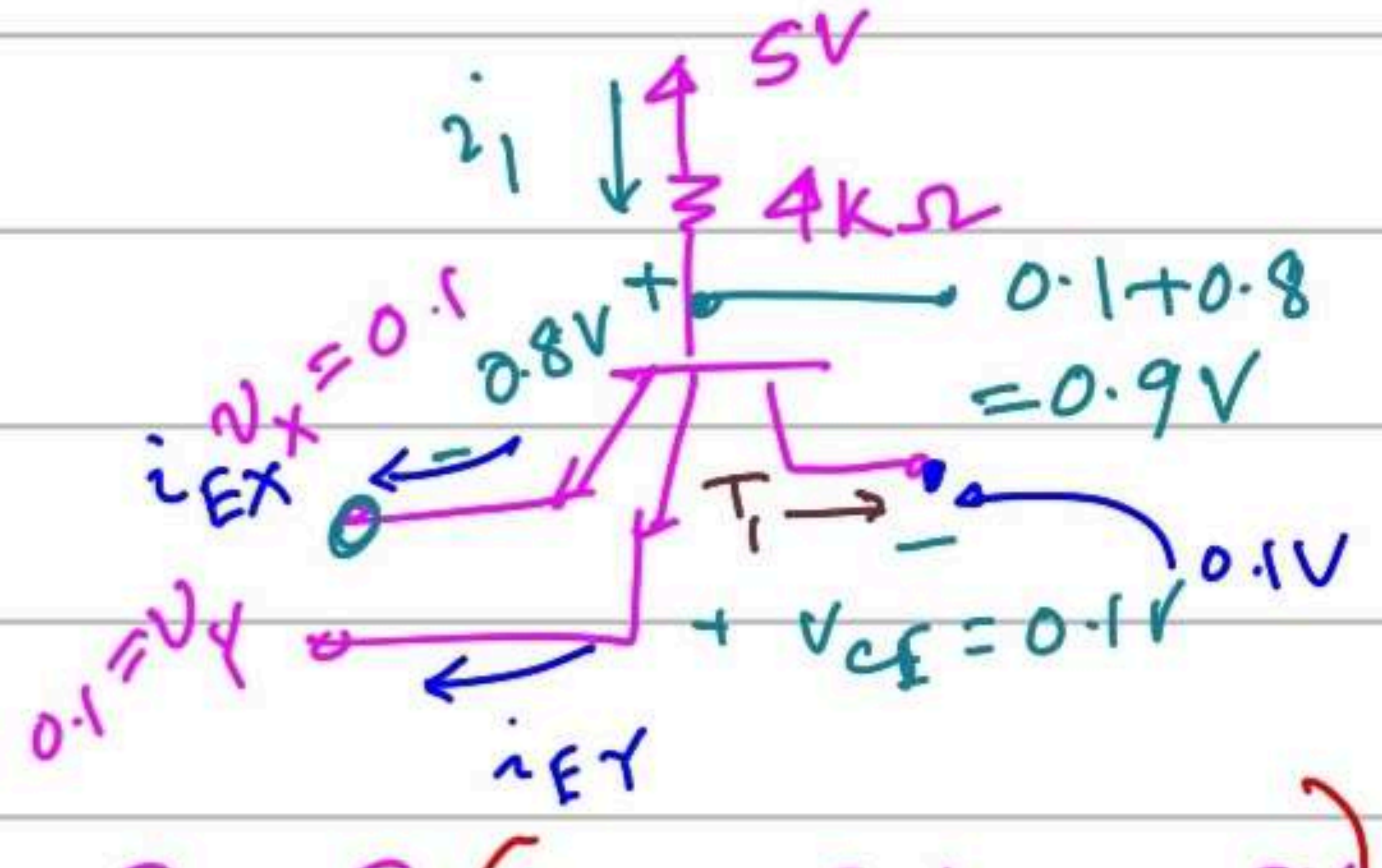
According to our assumption

$$i_2 = i_3 = i_R = i_{B0} = i_{E2} = i_{C1} = 0 \text{ mA}$$

Bonus Question: B-C terminal's bias ?

Ans: If B-C reverse bias, T_1 will be in forward active but for forward active we need. $i_C = \beta i_B$. But $i_C = 0$. Therefore not possible.

Hence T_1 must be in saturation mode. $\Rightarrow$ BC-junction is in forward bias

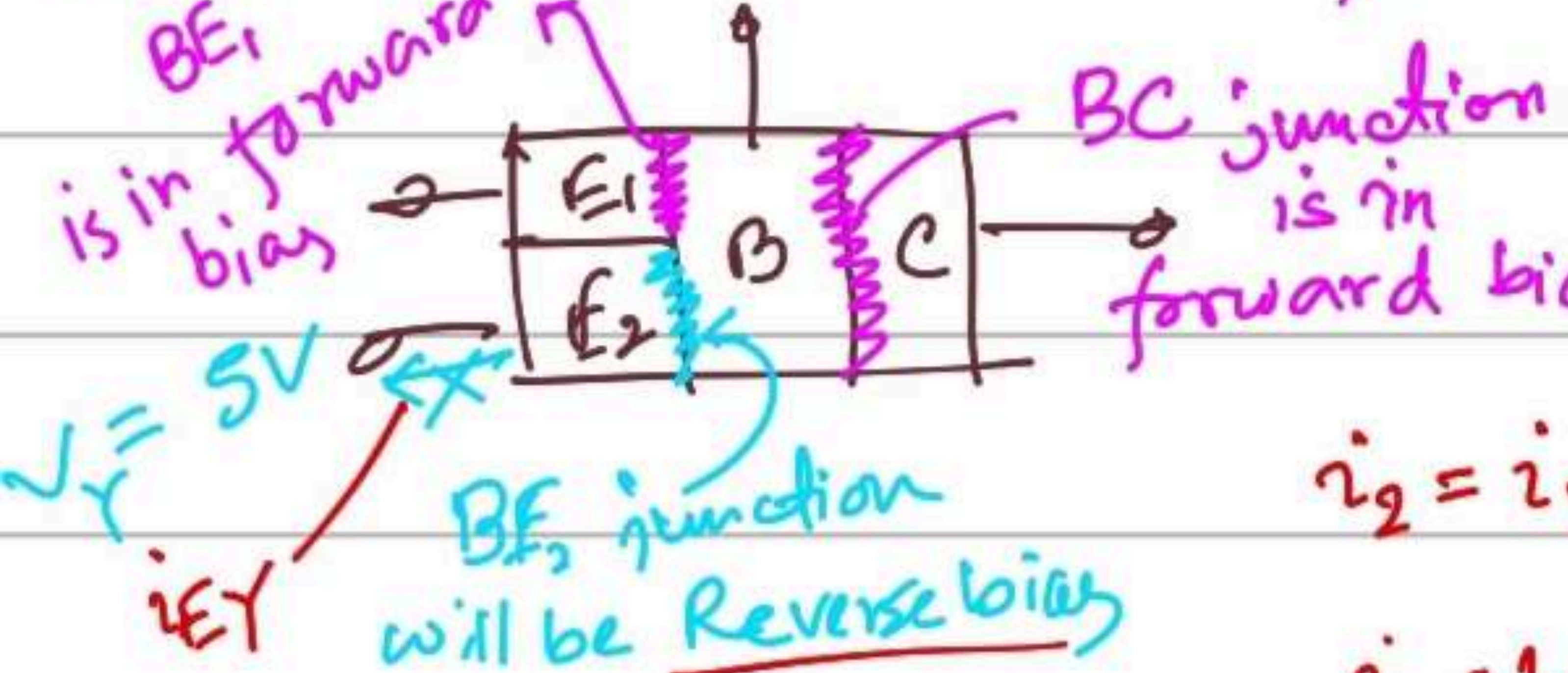


$$i_1 = \frac{5-0.9}{4K\Omega} = 1.125 \text{ mA}$$

$$i_{EX} = i_{EY} = \frac{i_1}{2} = 0.5675 \text{mA}$$

$\beta_{forced} = \frac{i_C}{i_B} = 0 < \beta_F$. Our assumption is valid.

Case ② $(v_X = 0.2V, \; V_Y = 5V)$ / $v_Y = 0.2V, \; v_X = 5V$



$T_1 \rightarrow$ saturation.

$T_o, T_2 \rightarrow$ cutoff.

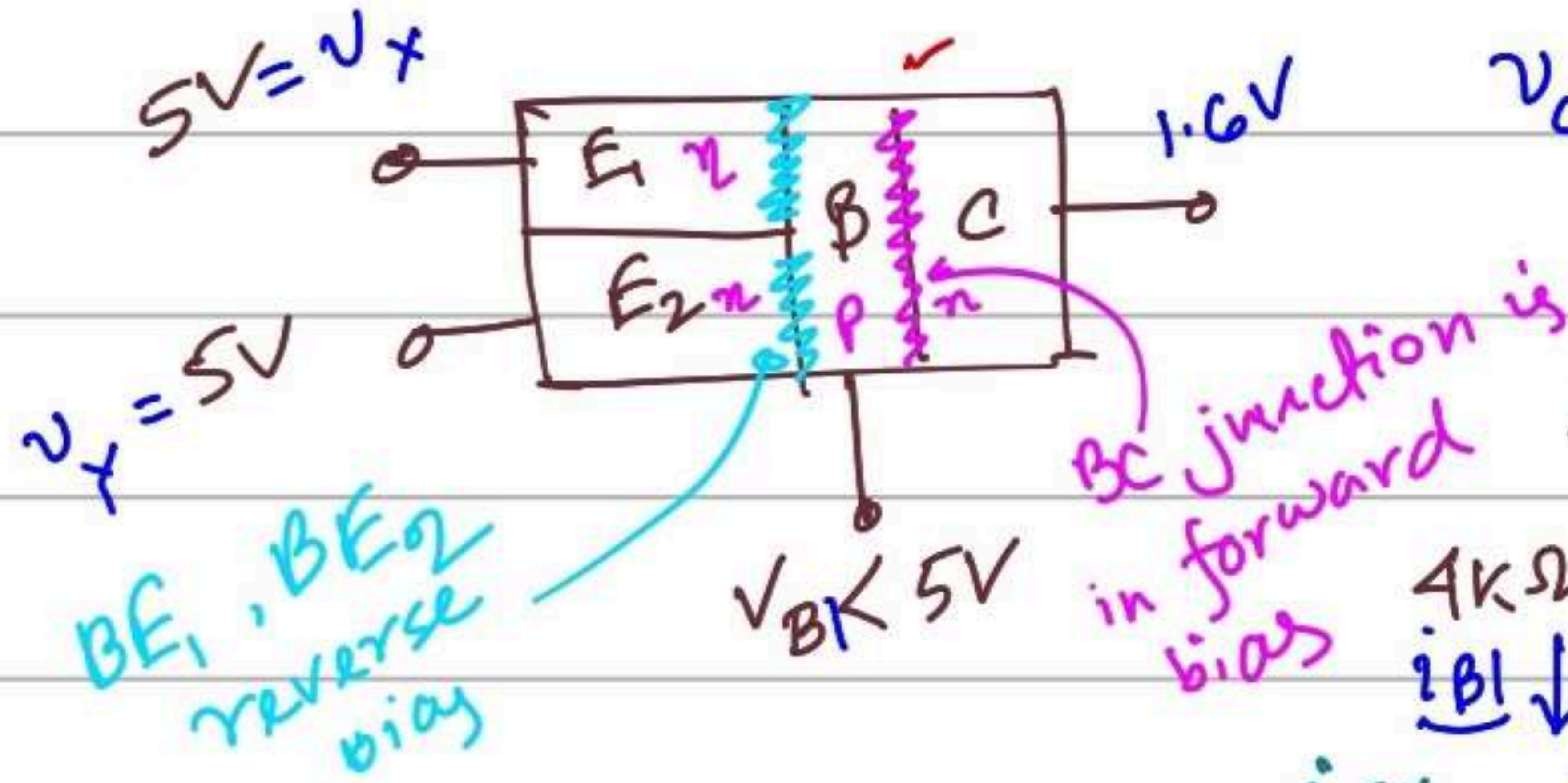
$$i_2 = i_3 = i_{B0} = i_{E2} = i_{C1} = i_R = 0$$

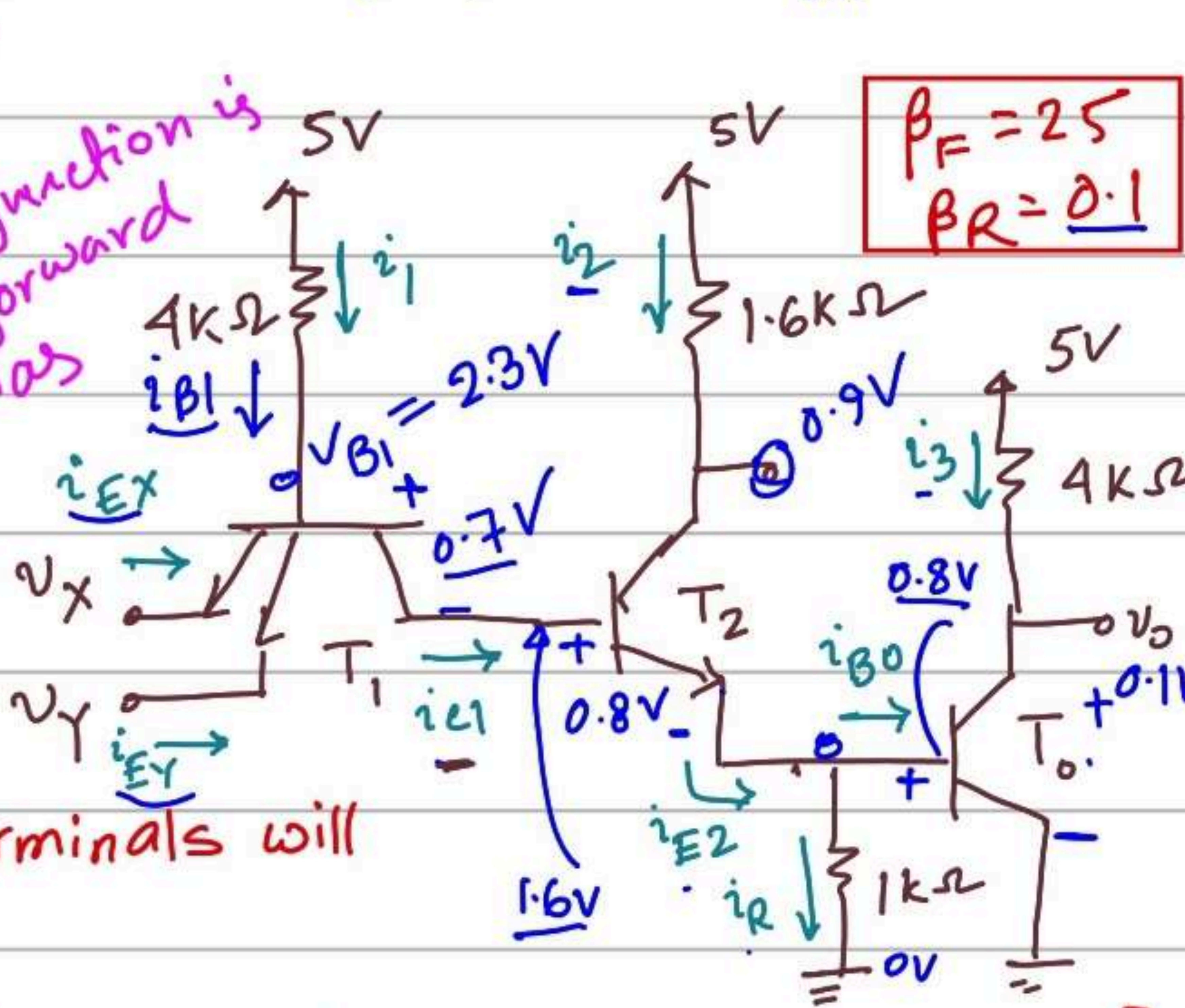
$i_1 = 1.125 \text{ mA}$. $\boxed{i_{EX} = i_1, \; i_{EY} = 0}$

Case 3: $v_X = v_Y = 5V$ | Assumption:

$T_0, T_2 \rightarrow$ saturation

$v_{CE}(sat) = 0.1\,V$, $v_{BE}(sat) = 0.8V$





\# T_1 transistor must be in reverse active mode.

\# Collector and Emitter terminals will switch their roles.

\# $V_{BC}\left(\text{reverse active}\right) = 0.7V$. [because BC junction is in forward bias]

\# $\left(i_{EX} = \beta_R i_B = i_{EY}\right)$ β_R = reverse active current gain.

$i_{B1} = i_1 = \frac{5-2.3}{4k} = 0.675\,mA$, $i_2 = \frac{5-0.9}{1.6k} = 2.5625\,mA$.

$i_3 = \frac{5-0.1}{4k} = 1.225\,mA$, $i_{EX} = i_{EY} = \beta_R i_{B1} = 0.0675\,mA$

$i_{C1} = i_{EX} + i_{EY} + i_{B1} = 0.81\,mA$, $i_{E2} = i_2 + i_{C1} = 3.3725\,mA$,

$i_R = \frac{0.8-0}{1k} = 0.8\,mA$, $i_{B0} = i_{E2} - i_R = 2.5725\,mA$

$\beta_{forced}(T_2) = \frac{i_2}{i_{C1}} = \frac{2.5625}{0.81} = 3.16 < \beta_F = 25$

$\beta_{forced}(T_0) = \frac{i_3}{i_{B0}} = \frac{1.225}{2.5725} \approx 0.47 < \beta_F = 25$

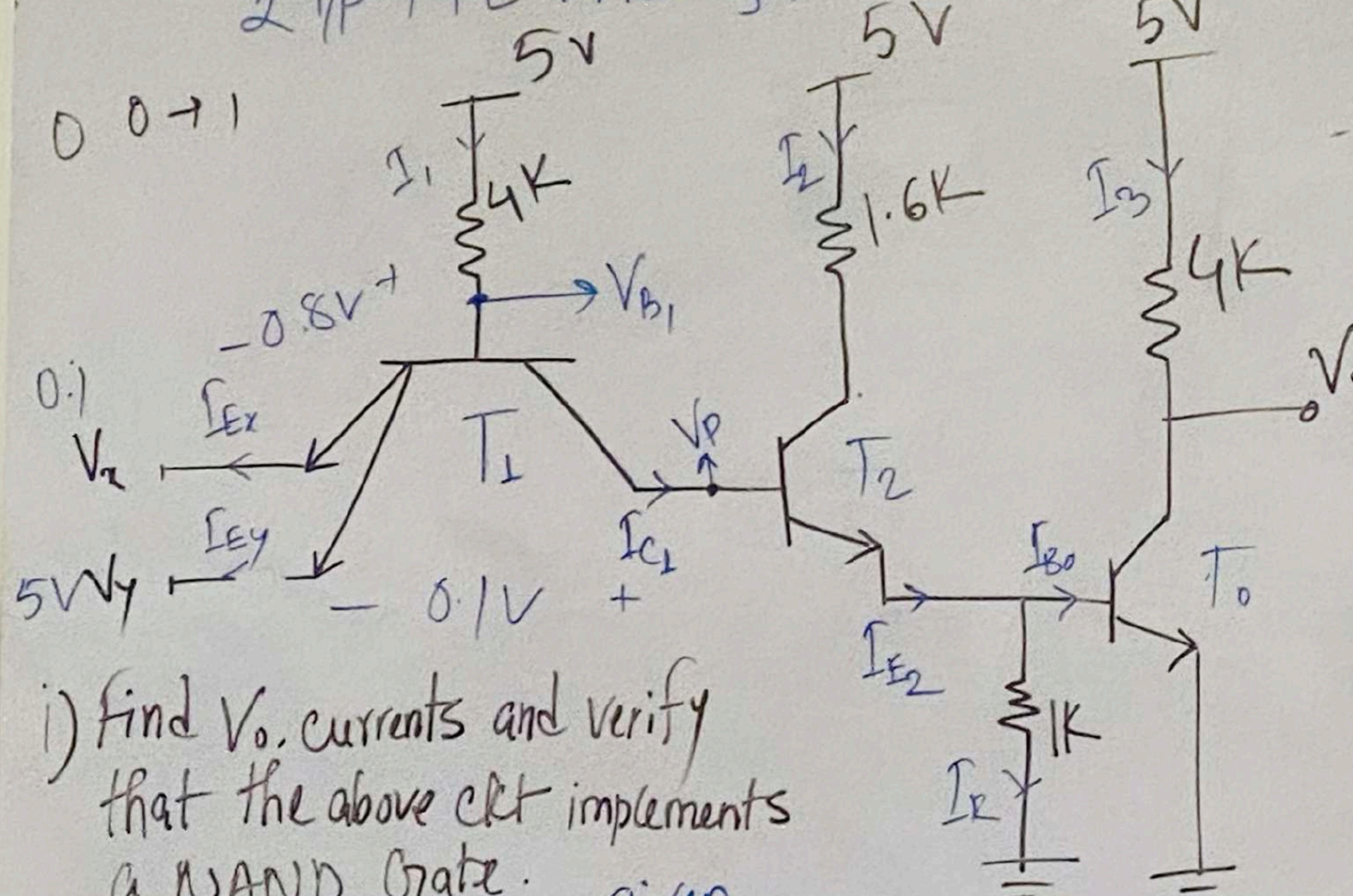
Our assumptions are valid.

v_X	v_Y	v_0
0.1	0.1	5
0.1	5	5
5	0.1	5
5	5	0.1

This circuit works like a NAND gate.

TTL (Transistor-Transistor logic)

2 i/p TTL NAND gate.



i) Find V_o, currents and verify that the above ckt implements a NAND Gate.

ii) Find P_T, P_{Avg}.

given

$V_{BE(sat)} = 0.8V$
$V_{CE(sat)} = 0.1V$

$V_{iL} = 0.1V$
$V_{iH} = 5V$

$\boxed{\beta_R = 0.1}$

Case-I

$V_x = 0.1V,\ V_y = 0.1V$

E_1B, E_2B Junction → FB

CB junction has to be → FB

$\therefore T_1 \rightarrow$ Saturation.

$\therefore V_{BE} = 0.8V,\quad V_{CE} = 0.1V$

$\therefore V_P = V_{CE} + V_x = 0.2V \rightarrow$

$\therefore T_2, T_o \rightarrow$ cutoff

$I_{e_1} = I_2 = I_{E2} = I_R = I_{B0} = I_3 = 0$

$I_3 = \frac{5 - V_o}{4K}$

$\Rightarrow 0 = 5 - V_o \rightarrow \boxed{V_o = 5V}$

$V_{B_1} = 0.8 + 0.1 = 0.9V$

$I_1 = \frac{5 - 0.9}{4K} = 1.025 mA$

$I_{Ex} = I_{Ey} = \frac{I_1}{2} = 0.512 mA$

P_1

$P_1 = (5-0.1)\times(I_{Ex}+I_{Ey})$

$= (5-0.1)\times I_1 = 5.0225\, mW$

Case-II

$V_x = 0.1V,\ V_y = 5\ V$

$E_1B \to FB$, E_2B junc $\to RB$

$\therefore CB$ junction $\to FB$

$I_1 \to$ saturation $\therefore V_P = 0.2V$

$\therefore T_2, T_0 \to$ cutoff

$\therefore I_{C_1} = I_2 = I_{E_2} = I_R = I_{B0} = I_3 = 0$

$\boxed{V_0 = 5V}$ $\boxed{V_{B_1} = 0.9V}$

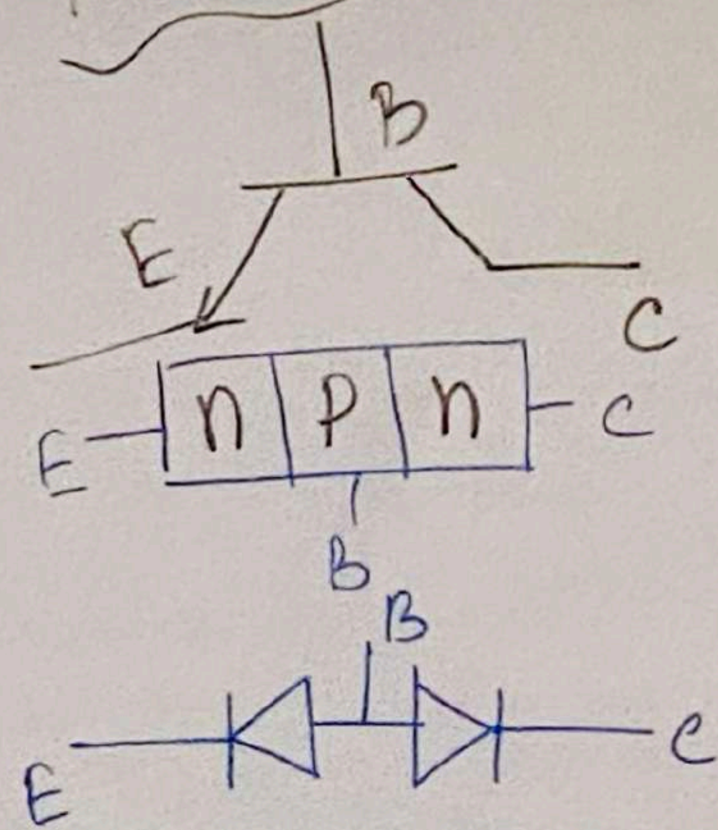
$I_1 = \frac{5-0.9}{4K} = 1.025$

$\boxed{I_{Ey} = 0}$ $\boxed{I_{Ex} = I_1}$

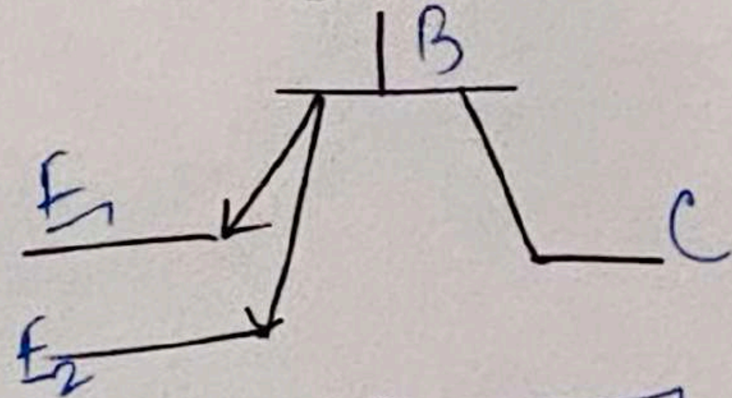
$P_2 = (5-0.1)\times I_1 = 5.0225\, mW$

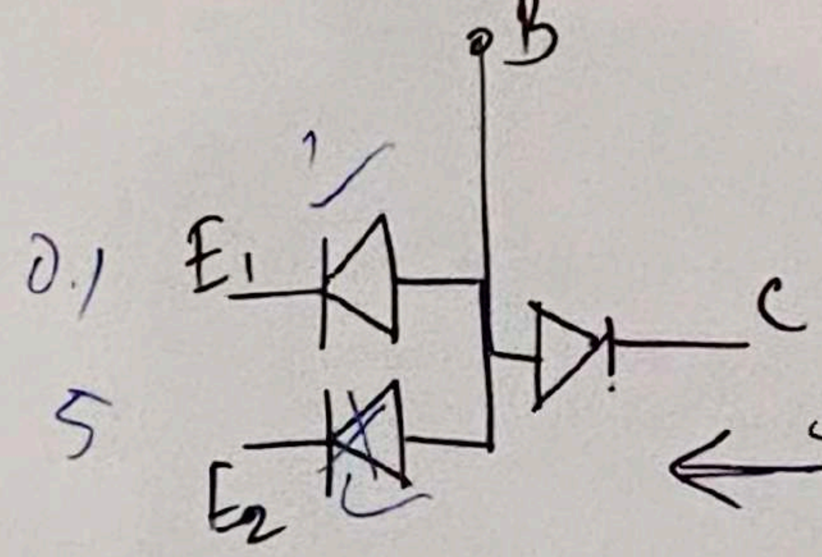
EBJ	C-BJ
FB	FB → Sat.
FB	RB → FA
RB	RB → cutoff
RB	FB → RA

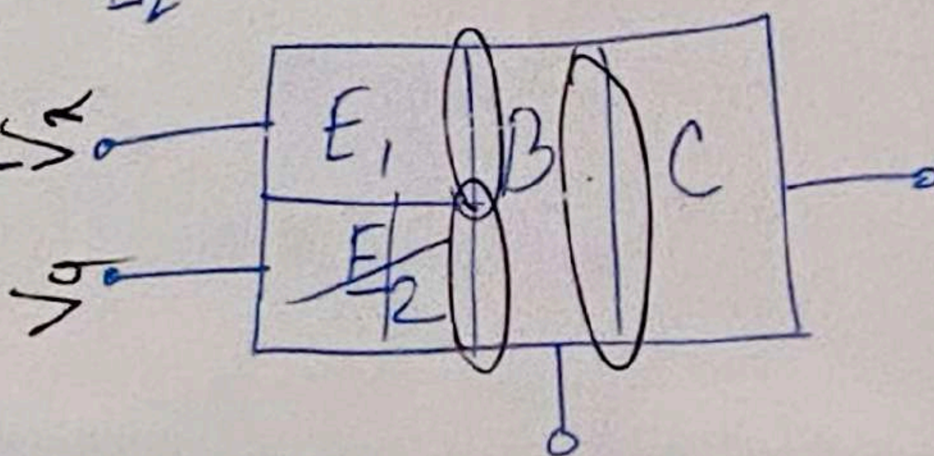
npn BJT



Multi-Emitter BJT



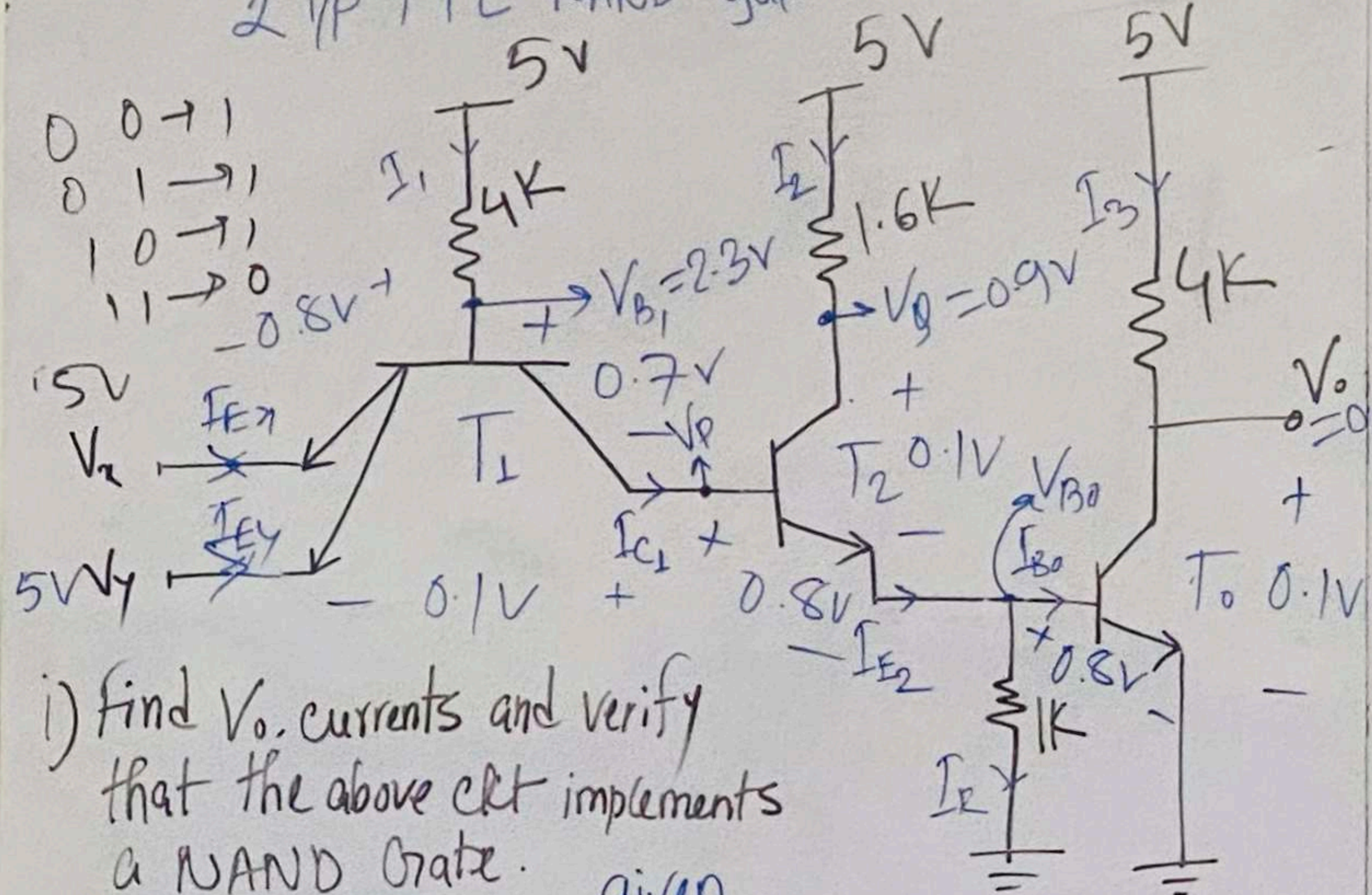




TTL (Transistor-Transistor logic)

2 i/p TTL NAND gate

0	0	→ 1
0	1	→ 1
1	0	→ 1
1	1	→ 0



i) Find V_o, currents and verify that the above ckt implements a NAND Gate.

ii) Find P_T, P_{Avg}.

given

$V_{BE(sat)} = 0.8V$

$V_{CE(sat)} = 0.1V$

$V_{iL} = 0.1V$

$V_{iH} = 5V$

$\boxed{\beta_R = 0.1}$

Case-III

Same as case II

$\boxed{P_3 = P_2}$

Case-IV

$V_x = 5V$, $V_y = 5V$

E_1B, E_2B junction → RB

CB junction → FB

T_1 → Reverse Active

T_2, T_o → Sat. → $V_{BE} = 0.8V$, $V_{CE} = 0.1V$

$\boxed{V_o = 0.1V}$ $\boxed{V_{B0} = 0.8V}$

$V_P = 0.8 + 0.8 = 1.6V$

$V_Q = 0.8 + 0.1 = 0.9V$

$V_{B_1} = 0.7 + 0.8 + 0.8 = 2.3V$

$$I_1 = \frac{5-2.3}{4K} = 0.67\,mA$$

$$I_2 = \frac{5-0.9}{1.6K} = 2.56\,mA$$

$$I_3 = \frac{5-0.1}{4K} = 1.225$$

$$\beta_R = \frac{I_E}{I_B} = \frac{I_{Ex}}{I_1}$$

$$\therefore I_{Ex} = \beta_R \times I_1 = 0.1 \times 0.67 = 0.067\,mA$$

$$I_{Ey} = \beta_R \times I_1 = 0.067\,mA$$

$$I_{E_1} = I_1 + I_{Ex} + I_{Ey} = 0.804\,mA$$

$$I_{E_2} = I_{E_1} + I_2 = 3.364\,mA$$

$$I_R = \frac{0.8-0}{1} = 0.8\,mA$$

$$I_{B0} = I_{E_2} - I_R = 2.564$$

3V

$$P_4 = (5-0) \times (I_{Ex} + I_{Ey} + I_1 + I_2 + I_3)$$

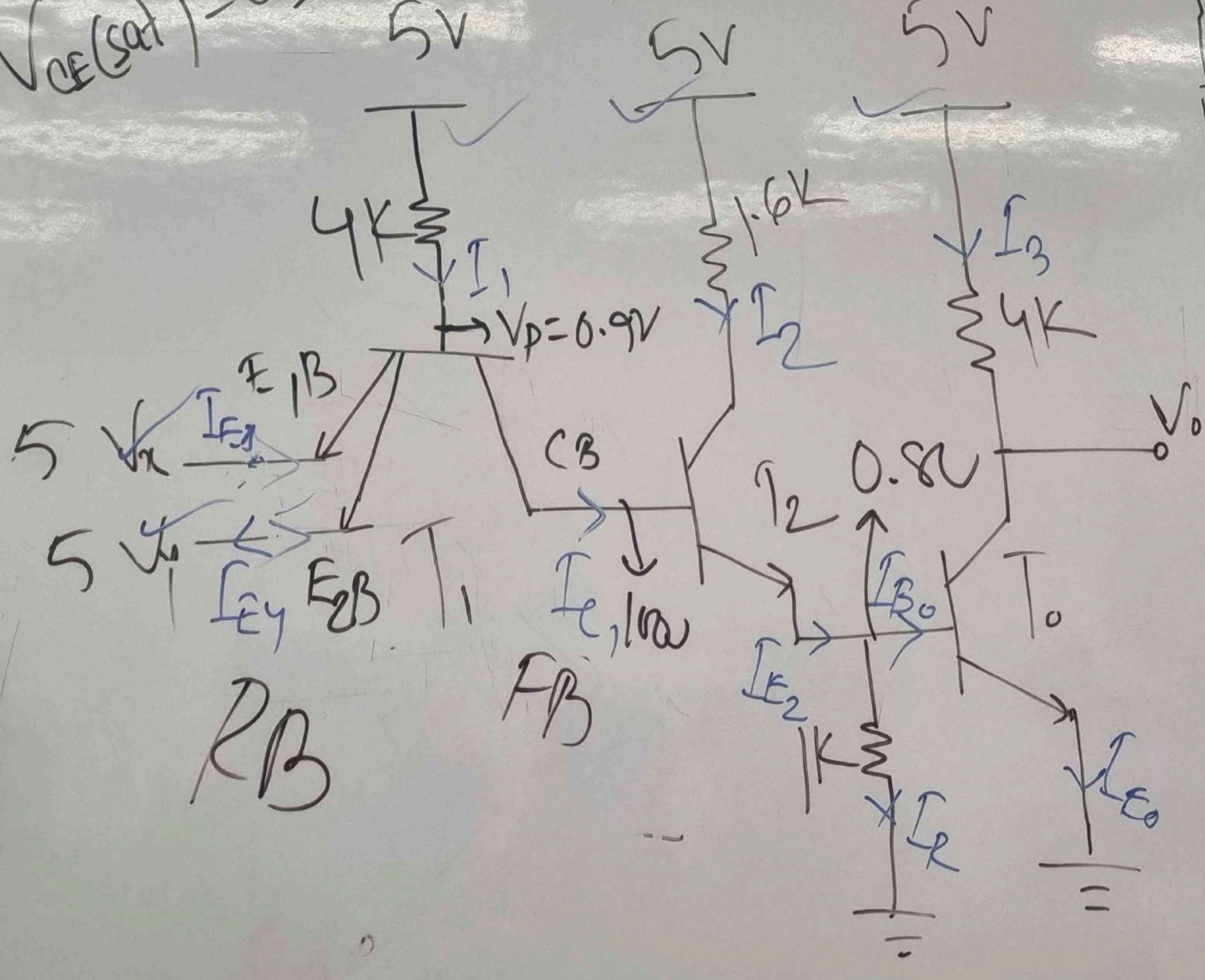
$$= 22.92\,mW$$

$$P_T = P_1 + P_2 + P_3 + P_4$$

$$P_{AVG} = \frac{P_T}{4}$$

Power Consumption of TTL

$V_{CE(sat)} = 0.1\,V$



V_x	V_y	V_0
0.1	0.1	5
0.1	5	5
5	0.1	5
5	5	0.1

NAND

Case-I

$V_x = 0$

$I_1 =$

0.1V

P_1

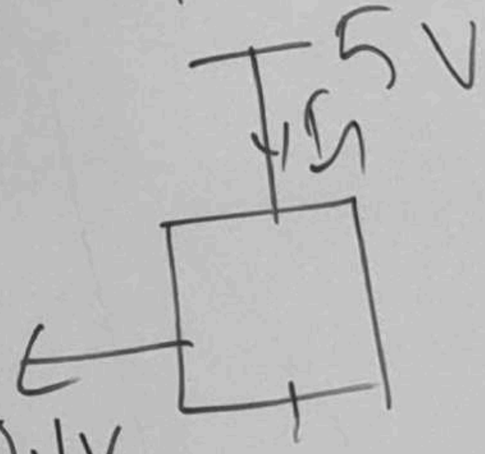
V_x	V_y	V_o
0.1	0.1	5
0.1	5	5
5	0.1	5
5	5	0.1

NAND

Case-I

$V_x = 0.1V, V_y = 0.1V$

$I_1 = 1.025\ mA$



$P_1 = (5-0.1) \times I_1$

$= 4.9 \times 1.025$

$= 5.0225\ mW$

Case-II

$V_x = 0.1V, V_y = 5V$

$I_1 = \frac{5-0.9}{4}$

$= 1.025\ mA$

$P_2 = (5-0.1) I_1$

$= 5.0225\ mW$

Case-III

$V_x = 5V, V_y = 0.1V$

Similarly $\rightarrow P_3 = 5.025\ mW$

Case-IV

$V_x = 5V, V_y = 5V$

$I_{Ex} = 0.0675$

$I_{Ey} = 0.0675$

$I_1 = 0.675, I_2 = 2.56, I_3 = 1.23\ mA$

$P_4 = (5-0)(I_x + I_y + I_1 + I_2 + I_3)$

$= 23\ mW$

Total $\rightarrow P_T = P_1 + P_2 + P_3 + P_4 = 38.06\ mW$

Avg $\rightarrow P_{Avg} = \frac{P_T}{4} = 9.51\ mW$